PROJECT SYNOPSIS

MOTORSPORTS TELEMETRY

Real-Time Telemetry Analytics and Predictive Modeling for Motorsports Strategy Optimization
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Introduction

Formula 1 racing represents the pinnacle of automotive technology and data-driven decision making. Modern F1 cars generate over 500 gigabytes of telemetry data per race weekend, transmitted at frequencies exceeding 100 data points per second. This massive data stream contains critical information about vehicle performance, tyre degradation, fuel consumption, and driver inputs that directly influence race outcomes. However, the challenge lies not merely in collecting this data, but in processing it in real-time, storing it efficiently, and extracting actionable insights that can inform split-second strategic decisions.

Problem Statement

Formula 1 racing generates massive amounts of telemetry data, but analyzing this data for race strategy decisions requires expensive proprietary systems costing hundreds of thousands of dollars. Current solutions lack integration of real-time data capture, efficient storage, and machine learning-based predictions. Teams need accessible platforms that can process high-frequency telemetry streams, store data efficiently without consuming excessive storage space, and predict lap times based on race conditions.

Proposed Solution

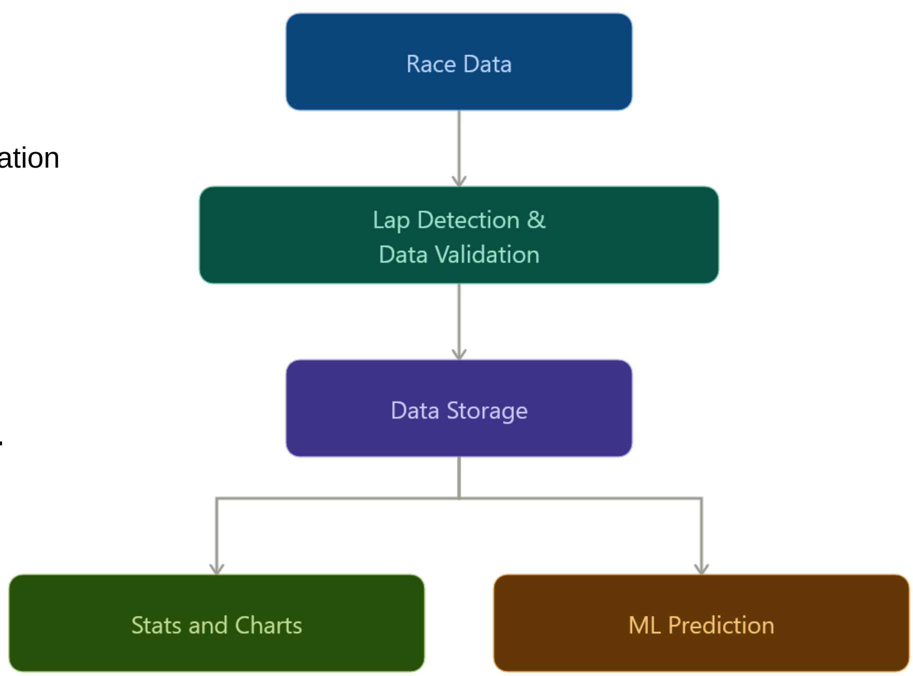
This project presents an integrated telemetry analytics platform that requires four primary layers: data capture, data storage, data analysis, and predictive modeling. The solution demonstrates a production-ready engineering practices while maintaining accessibility for educational and research purposes.

Solution Architecture:

- Layer 1: Real-Time Data Capture
- Layer 2: Persistent Data Storage
- Layer 3: Statistical Analysis and Visualization
- Layer 4: Predictive Machine Learning

Additional Components:

- Web Dashboard (Flask-based)
- User Interface Layer



Key Features & Capabilities

- Intelligent Storage Optimization
- Real-Time Data Capture
- Comprehensive Performance Analysis
- Advanced ML Capabilities
- Interactive Visualization Dashboard
- Strategy Event Detection

Technology Stack

CATEGORY	TOOLS & TECHNOLOGIES
Programming Language	Python
Core Libraries	Socket, struct, mysql-connector-python, pandas, numpy, matplotlib, flash, joblib, scikit-learn, FastF1
Database Technology	MySQL
Frontend Technology	HTML, CSS, JavaScript, Chart,js
Data Source	FastF1, F12018 (Codemasters)

Expected Outcomes

- Real Time Data Capture
- Comprehensive Performance Analysis
- Advanced ML Capabilities
- Interactive Visualization Dashboard
- Strategy Event Detection
- Dashboard Connection

Project Scope & Limitations

In Scope: Data Capture, Data Storage, Data Analysis, Machine Learning, Visualization

Current Limitations:

- Telemetry Data Quality
- Model Training Constraints
- Database Schema Limitations
- User Experience
- Deployment and Scale Limitations
- Dashboard Connection

Conclusion

This F1 Telemetry and Strategy Analytics Platform represents a comprehensive demonstration of modern data engineering and machine learning practices, successfully addressing the challenge of building production-ready analytics systems from real-time data sources. The project achieves its core objectives: capturing high-frequency telemetry at 60 packets per second with zero data loss, implementing efficient storage strategies reducing database footprint by 98%, and deploying machine learning models achieving 91% predictive accuracy with mean absolute error of just 1.4 seconds.